See It, Say It, Pick It

Track 2A technical report
Team: Nightcats

1. System

The system converts one workcell image and one English instruction into a checked robot plan and first-pick pixel. Apertus v1.5 8B performs the language reasoning. Vision models supply object candidates and masks, while deterministic checks decide whether to plan, ask for clarification or refuse.

2. Where Apertus had difficulty

Compositional instructions. Short literal commands were reliable, but nested relations, coordinated clauses, negation and scope were less stable.

Structured output. Free-form generation sometimes returned null instead of an empty list, omitted one half of a relation pair, or dropped a qualifier such as a color or destination.

Language-perception mismatch. Apertus may correctly refer to a gear while the vision inventory calls the same object a sun or `graspable_object`. The language output is then valid, but the resolver cannot bind it to the scene.

3. How we handled it

Apertus produces a small typed intent rather than a free-form robot action. We split coordinated instructions into clauses, constrain the output schema, verify that every source qualifier is preserved and allow one bounded semantic repair. A deterministic resolver then checks identity, relations, uniqueness, feasibility and safety. Apertus never invents pixels or overrides these checks.

4. Evaluation

The official public development-set score was **24.0/65**: grounding ratio **0.30**, safety contribution ratio **0.45**, unsafe or impossible catch rate **0.90**, and false-alarm rate **0.50**. The system catches most unsafe cases, but conservative resolution rejects too many valid tasks.

5. Remaining limitations

Open-vocabulary matching remains the main weakness. Generic labels, duplicate detections, conflicting masks and visually similar containers still create missing or ambiguous references. The next step is model-backed semantic matching, followed by duplicate suppression and stronger evidence for grasp and container feasibility.